

62) How many of the disjunctions $p \vee \neg q$, $\neg p \vee q$, $q \vee r$, $q \vee \neg r$ and $\neg q \vee \neg r$ can be made simultaneously true by an assignment of truth values to p , q and r ?

Soln:

p	q	r	$p \vee \neg q$	$\neg p \vee q$	$q \vee r$	$q \vee \neg r$	$\neg q \vee \neg r$
T	T	T	T	T	T	T	F
T	T	F	T	T	T	F	T
T	F	T	T	F	T	T	T
T	F	F	T	F	F	T	F
F	T	T	F	T	T	T	F
F	T	F	F	T	T	T	T
F	F	T	T	T	F	T	T
F	F	F	T	T	F	T	T

→ All 5 are true at the same time